

UNIVERSITY OF CLAUDE · FREE RESOURCE MAP

Resource Map

CSCE 633 Machine Learning — module-by-module mapping onto free university courseware

HOW TO USE THIS

This is your lecture hall. For each module: watch or read the mapped external material first, then work the module slides and notes, which are written to assume you have seen it.

Where a specific lecture number is given, that is the one that covers this module. Where a book chapter is given, read that chapter, not the whole book.

Everything here is free. Resources remain under their own licences — link to them, do not redistribute them.

Primary sources for the whole course

Resource	Role in this course
Stanford CS229 — Machine Learning (free lectures and notes)	The primary source. Andrew Ng's lecture notes are the best free derivation-level treatment of Modules 02 through 08, and the problem sets are worth doing.
James, Witten, Hastie & Tibshirani — An Introduction to Statistical Learning (free PDF)	Free, and the right level for this course. Strong on the bias–variance material of Module 03 and on resampling. The R and Python editions are both free.
Hastie, Tibshirani & Friedman — The Elements of Statistical Learning (free PDF)	The deeper companion to the above, also free. Use it for the derivations the first book states without proof.
scikit-learn — the user guide (free)	Unusually good documentation — the model selection and metrics chapters are genuinely instructional, not merely reference material.
Google — Rules of Machine Learning (free)	The Module 13 material. Forty-three rules from people who have deployed a great deal of this, and the early ones are about not using machine learning yet.
Kaufman, Rosset & Perlich — Leakage in Data Mining (free)	The Module 12 reference. Worked examples of evaluations that were wrong and the competitions they won.

Module-by-module map

Module 01 — What Learning From Data Means

Resource	What it covers for this module
Stanford CS229 — lecture 1 and the supervised learning notes (free)	The §1 formulation, derived rather than asserted.
James et al. — An Introduction to Statistical Learning, chapter 2 (free PDF)	The best free statement of §1 and §3 , with the accuracy-versus-interpretability trade laid out honestly.
Google — Rules of Machine Learning (free)	The §4 material. Rules 1 through 3 are about not using machine learning yet, which is the most useful advice in the document.
Kapoor & Narayanan — Leakage and the Reproducibility Crisis in ML-based Science (free)	The §2 callout, with a survey of how often it happens — across seventeen fields, and the answer is 'constantly'.

Module 02 — Linear Models

Resource	What it covers for this module
Stanford CS229 — supervised learning notes, parts I and II (free)	The derivations of §1 and §3 , including the generalised linear model framing that explains why the two gradients match.
James et al. — An Introduction to Statistical Learning, chapters 3 and 4 (free PDF)	Linear and logistic regression with the statistical interpretation of the coefficients, which §4 relies on.
Sebastian Ruder — An Overview of Gradient Descent Optimization Algorithms (free)	The §2 variants, compared clearly, with the update rules written out.
Trefethen & Bau — Numerical Linear Algebra	Why not to form $\mathbf{X}^T\mathbf{X}$ — the conditioning argument of §1, done properly. Lecture 11 is the relevant one.

Module 03 — Bias, Variance, and Capacity

Resource	What it covers for this module
James et al. — An Introduction to Statistical Learning, chapter 2 (free PDF)	The decomposition of §1 , derived, with the clearest available figures.
Andrew Ng — Machine Learning Yearning (free)	The §2 diagnostic procedure as an engineering discipline — short chapters, entirely about deciding what to do next.
Belkin, Hsu, Ma & Mandal — Reconciling modern machine learning practice and the bias-variance trade-off (free)	The double descent paper. The figures are the argument.
Hastie et al. — The Elements of Statistical Learning, chapter 7 (free PDF)	Model assessment and selection at depth, including the effective degrees of freedom notion behind §3.

Module 04 — Regularisation and Model Selection

Resource	What it covers for this module
Hastie et al. — The Elements of Statistical Learning, chapters 3 and 7 (free PDF)	Ridge and lasso with the geometric argument of §1 , and the model selection theory of §2.
scikit-learn — cross-validation and model selection user guide (free)	The §2 pipeline discipline , with the leakage warning stated explicitly in the documentation — which is unusual and welcome.
Bergstra & Bengio — Random Search for Hyper-Parameter Optimization (free)	The §3 result . The figures showing why grid search wastes its budget are the whole argument.
Cawley & Talbot — On Over-fitting in Model Selection and Subsequent Selection Bias (free)	Why the inner-loop score is optimistic — the nested CV argument of §2, measured.

Module 05 — Evaluation and Metrics

Resource	What it covers for this module
Saito & Rehmsmeier — The Precision-Recall Plot Is More Informative than the ROC Plot (free)	The §2 argument , with worked examples at several prevalences. Short and conclusive.
Guo, Pleiss, Sun & Weinberger — On Calibration of Modern Neural Networks (free)	The §3 result . Modern networks are far less calibrated than older ones, measured and explained.
scikit-learn — metrics and probability calibration guides (free)	The implementations, with unusually good discussion of when each metric is appropriate.
Blagec et al.; and the imbalanced-learn documentation (free)	The §4 resampling pitfalls , including the resample-before-split error, documented by the library that implements the methods.

Module 06 — Trees and Ensembles

Resource	What it covers for this module
Hastie et al. — The Elements of Statistical Learning, chapters 9, 10 and 15 (free PDF)	Trees, boosting, and random forests, with the bias-versus-variance framing of §2 and §3 made precise.
Friedman — Greedy Function Approximation: A Gradient Boosting Machine (free)	The §3 algorithm, from its author . The gradient framing is what makes it general.
Grinsztajn, Oyallon & Varoquaux — Why do tree-based models still outperform deep learning on tabular data? (free)	The §3 callout, measured carefully across many datasets, with the reasons isolated experimentally.
Strobl et al. — Bias in random forest variable importance measures (free)	The high-cardinality bias of §4 , demonstrated. Read before reporting a Gini importance ranking.

Module 07 — Kernels and Margins

Resource	What it covers for this module
Stanford CS229 — support vector machines notes (free)	The derivation of §1 and §2 , including the dual formulation that makes the kernel substitution visible.
Hastie et al. — The Elements of Statistical Learning, chapter 12 (free PDF)	SVMs with the regularisation framing, which makes the C - λ connection of §1 explicit.
Schölkopf & Smola — Learning with Kernels	The reference for the representer theorem and for constructing custom kernels (§2). Library copy.
scikit-learn — SVM user guide and the kernel approximation module (free)	Practical tuning of C and γ , and the Nyström and random feature approximations that address §3's scaling problem.

Module 08 — Clustering and Density

Resource	What it covers for this module
Hastie et al. — The Elements of Statistical Learning, chapter 14 (free PDF)	Clustering, EM, and the validation difficulty of §3, treated with appropriate scepticism.
Arthur & Vassilvitskii — k-means++: The Advantages of Careful Seeding (free)	The §1 initialisation, with the approximation guarantee.
Campello, Moulavi & Sander — HDBSCAN (free), and the documentation	The §2 method that supersedes DBSCAN , with unusually good documentation about when clustering is and is not appropriate.
von Luxburg, Williamson & Guyon — Clustering: Science or Art? (free)	The §3 and §4 argument, made properly — why clustering resists objective evaluation and what follows.

Module 09 — Dimensionality Reduction

Resource	What it covers for this module
Hastie et al. — The Elements of Statistical Learning, chapter 14 (free PDF)	PCA with all three derivations of §1, and the manifold methods.
Wattenberg, Viégas & Johnson — How to Use t-SNE Effectively (free, interactive)	The §3 table, made interactive. Work through it — seeing t-SNE produce clusters from noise yourself is what makes the caveats stick.
McInnes, Healy & Melville — UMAP (free), and the documentation	The method, and an unusually honest section on what the plots do not show.
Johnson & Lindenstrauss; and scikit-learn's random projection module (free)	The §4 alternative, with the distortion bound and the dimension estimate.

Module 10 — Probability and Uncertainty

Resource	What it covers for this module
Murphy — Probabilistic Machine Learning: An Introduction (free PDF)	Free in full. The MLE/MAP/Bayes material of §1 and the uncertainty treatment of §3, done properly.
Rennie et al. — Tackling the Poor Assumptions of Naive Bayes Text Classifiers (free)	Why §2's assumption survives being false , and what to fix when it does not.
Kendall & Gal — What Uncertainties Do We Need in Bayesian Deep Learning? (free)	The aleatoric/epistemic distinction of §3 , with methods for estimating each.
Angelopoulos & Bates — A Gentle Introduction to Conformal Prediction (free)	The §4 method , explained without measure theory. Short, and it should be far better known.

Module 11 — Features, Data, and Labels

Resource	What it covers for this module
Northcutt, Athalye & Mueller — Pervasive Label Errors in Test Sets (free)	The §4 audit. Ten benchmarks, the error rates, and the ranking changes. Read it before trusting any benchmark number.
Andrew Ng — Machine Learning Yearning, and the data-centric AI material (free)	The §1 argument and the error-analysis procedure , stated as an engineering discipline.
Zheng & Casari — Feature Engineering for Machine Learning	The §2 and §3 catalogue, with the reasoning for each transform.
Ratner et al. — Snorkel: Rapid Training Data Creation with Weak Supervision (free)	The §4 weak supervision approach, with the accuracy-modelling machinery.

Module 12 — How Evaluations Lie

Resource	What it covers for this module
Kaufman, Rosset & Perlich — Leakage in Data Mining (free)	The §1 taxonomy , with competition-winning examples that were leakage.
Kapoor & Narayanan — Leakage and the Reproducibility Crisis in ML-based Science (free)	How often §1 happens — surveyed across seventeen fields, with a checklist worth adopting.
Recht, Roelofs, Schmidt & Shamir — Do ImageNet Classifiers Generalize to ImageNet? (free)	The §2 benchmark-drift measurement. New test sets, collected the same way, and the accuracy drop.
Dwork et al. — The reusable holdout (free)	The adaptivity problem of §2, with a principled method for reusing a holdout set that bounds the damage.

Module 13 — Deployment

Resource	What it covers for this module
Sculley et al. — Hidden Technical Debt in Machine Learning Systems (free)	The deployment reality of §1 and §2 , from Google. Feedback loops, entanglement, and why the model is the small part.
Google — Rules of Machine Learning, and the Model Cards paper (free)	The operational practices of §1, and the documentation artifact of §3.
Kleinberg, Mullainathan & Raghavan — Inherent Trade-Offs in the Fair Determination of Risk Scores (free)	The §3 impossibility theorem. Read the statement even if you skip the proof — it reframes the whole debate.
Barocas, Hardt & Narayanan — Fairness and Machine Learning (free book)	Free in full , and the best treatment of §3 as an engineering and measurement problem rather than a slogan.